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Aim:

Data Ingestion and Cleaning: Acquire data and prepare it for analysis.

Theory

Q1. what is data ingestion and why is it important in data science ? Which python library is commonly used for data ingestion and cleaning ? Ans: Data ingestion is the process of collecting, importing and transferring data from various source into a system where it can be a stored , processed and analyzed. These source can include :

1. Databases
2. CSV/Excel files
3. APIs 4.Web scraping 5.Streaming data Data ingestion can be : 6.Batch ingestion - Data is collected and processed in chunks at scheduled intervals . 7.Real-time ingestion -Data is continuously imported as it is generated . Importance in data science : Data ingestion is critical because : 8.Foundation of analysis -data science begins with data .Without properly ingested data analysis cannot happen . 9.ensure data availability - It makes raw data accessible. Q2. Why is data cleaning is necessary before performing analysis or machine learning? Ans: Data cleaning is necessary because raw data often contains errors , missing values , duplicates or inconsistencies . if not cleaned , it can be lead to :
4. incorrect analysis result 2.poor machine learning model performance . 3.biased or misleading prediction. Q3. How do you load CSV files into a pandas Data frame ? Ans : To load CSV file into pandas we use read_csv() function . syntax: df =pd.read_csv('your_file.csv') print(df.head()) common useful parameters : sep: change the delimiter header : Define which row to use as column names . usecols : Load only specific column to save memory. nrows: Only load the first n rows of a massive file Q4. What is the difference between head() and tail() function in pandas ? Ans : head() : 1.return the first n row of the dataset. 2.Use this to check column headers , data types and ensures the first few rows align with your expectation. tail(): 3.Returns the first n rows of the data set
5. Use this to check the total height of your dataset or to see if there are empty rows at the very end of a file . Q5. How can you identify missing values in a dataset? Ans: Missing values can be identified by checking for null , Nan , blank cells or special symbols in the data set. Tis can be done using function such as isnull() in python , ISBLANK() or COUNTBLANK() in Excel, or is NULL in SQL to detect and count missing data

```
from google.colab
import files
# Upload the data.csv file
uploaded = files.upload()
# You should see 'data.csv'
listed
as one of the uploaded files
```

UBprlowadse w.i.dget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable. Saving data.csv.xlsx to data.csv.xlsx

```
import pandas as pd
# 1. Load Excel file
df = pd.read_excel("data.csv.xlsx")
# Display first five rows
print("First 5 rows:")
print(df.head())
# 2. Identify missing values
print("\nMissing values:")
print(df.isnull())
# Count missing values
print("\nMissing values count:")
print(df.isnull().sum())
# 3. Replace missing values with 0
df_filled = df.fillna(0)
print("\nAfter replacing missing values with 0:")
print(df_filled)
# 4. Rename column names to lowercase
df_filled.columns = df_filled.columns.str.lower()
print("\nColumn names in lowercase:")
print(df_filled.columns)
# 5. Display dataset shape
```

```
print("\nDataset shape (rows, columns):")
print(df_filled.shape)
```

```
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FileNotFoundError                                Traceback (most recent call last)
/tmp/ipykernel_2559/3458258544.py in <cell line: 0>()
      1 import pandas as pd
      2 # 1. Load Excel file
----> 3 df = pd.read_excel("data.csv.xlsx")
      4 # Display first five rows
      5 print("First 5 rows:")
```

```
----- 3 frames -----
/usr/local/lib/python3.12/dist-packages/pandas/io/common.py in get_handle(path_or_buf, mode, encoding, compression,
memory_map, is_text, errors, storage_options)
      880         else:
      881             # Binary mode
--> 882             handle = open(handle, ioargs.mode)
      883             handles.append(handle)
      884
```

```
FileNotFoundError: [Errno 2] No such file or directory: 'data.csv.xlsx'
```

```
print('Files in the current Colab environment:')
!ls -F
```

```
Files in the current Colab environment:
sample_data/
```